

Lizhi (Gary) Yang

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Research Interests

My research aims to create robust, safe, and intelligent robots that can operate reliably in the real world. I focus on the intersection of robot safety, reinforcement learning, perception-enabled control, and loco-manipulation.

Education

California Institute of Technology

Ph.D. Candidate, Mechanical Engineering (Robotics, Controls and Dynamics)

Pasadena, CA

Sept 2022 – Present

California Institute of Technology

M.S. Mechanical Engineering (along the way)

Pasadena, CA

June 2024

University of California, Berkeley

B.S. Electrical Engineering and Computer Sciences

Berkeley, CA

May 2022

Professional Experience

California Institute of Technology

Ph.D. Candidate

Pasadena, CA

2022 – Current

- Built full-stack pipeline of humanoid robot whole-body task-space controller, enabling the Unitree G1 to complete natural language and vision driven tasks in the real world with no task data collection.
- Constructed social-semantic safety frameworks for general mobile robot platforms, validated on the Unitree G1 and Go2.
- Developed pipelines for data collection, finetuning, and deployment of VLA models for Franka Emika.
- Designed reinforcement learning and hardware deployment pipeline for humanoid robots, enabling real-world experiments.
- Developed methods to enforce safety in learned control and planning policies, improving robustness of autonomous systems.
- Created push-recovery and uncertainty-aware locomotion frameworks, validated on humanoid platforms.

University of California, Berkeley

Student Research Assistant

Berkeley, CA

2019 – 2022

- Conducted robotics research on locomotion, perception, and multi-robot collaboration, with deployments on Unitree A1 and MIT Mini Cheetah.
- Built multi-robot system frameworks integrating LiDAR and depth sensing for collaborative load-moving tasks.
- Developed vision-based navigation and obstacle detection stacks in ROS, enabling autonomous quadrupedal navigation.
- Applied reinforcement learning (PPO) and Bayesian optimization for locomotion control, improving stability and adaptability.

Samsara Inc.

Software Engineering Intern

San Francisco, CA

2021

- Delivered production features in a 4-person engineering team, contributing to fleet management products.
- Built machine learning benchmarking tools to compare models at scale, improving workflow efficiency for ML engineers.
- Leveraged C++ and Go for backend development and performance optimization.

Awards & Honors

2025: Best Oral Presentation Finalist, International Conference on Humanoid Robots (Humanoids)

2022: Arthur M. Hopkin Award – University of California, Berkeley

2021: Best Service Robotics Paper Finalist, International Conference on Robotics and Automation (ICRA)

Publications (* denotes equal contribution)

- [1] ***Lizhi Yang**, *Blake Werner, Massimiliano de Sa, and Aaron D. Ames. “CBF-RL: Safety Filtering Reinforcement Learning in Training with Control Barrier Functions”. In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2026.
- [2] Min Dai, William D. Compton, Junheng Li, **Lizhi Yang**, and Aaron D. Ames. “Walk the PLANC: Physics-Guided RL for Agile Humanoid Locomotion on Constrained Footholds”. In: *arXiv:2601.06286*. Preprint. 2026.
- [3] **Lizhi Yang**, Ryan M. Bena, Meg Wilkinson, Gilbert Bahati, Andy Navarro Brenes, Ryan K. Cosner, and Aaron D. Ames. “Safe-SAGE: Social-Semantic Adaptive Guidance for Safe Engagement through Laplace-Modulated Poisson Safety Functions”. In: *arXiv:2603.05497*. Preprint. 2026.

- [4] ***Lizhi Yang**, *Blake Werner, Ryan K. Cosner, David Fridovich-Keil, Preston Culbertson, and Aaron D. Ames. "SHIELD: Safety on Humanoids via CBFs In Expectation on Learned Dynamics". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025.
- [5] Ryan M. Bena, Gilbert Bahati, Blake Werner, Ryan K. Cosner, **Lizhi Yang**, and Aaron D. Ames. "Geometry-Aware Predictive Safety Filters on Humanoids: From Poisson Safety Functions to CBF-Constrained MPC". In: *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*. 2025.
- [6] **Lizhi Yang**, Blake Werner, Adrian B. Ghansah, and Aaron D. Ames. "Bracing for Impact: Robust Humanoid Push Recovery and Locomotion with Reduced Order Models". In: *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*. 2025.
- [7] Pol Mestres, Shima Sadat Mousavi, Pio Ong, **Lizhi Yang**, Ersin Das, Joel W. Burdick, and Aaron D. Ames. "Explicit Control Barrier Function-based Safety Filters and their Resource-Aware Computation". In: *arXiv:2512.10118*. Preprint. 2025.
- [8] *Blake Werner, ***Lizhi Yang**, and Aaron D. Ames. "Architecture Is All You Need: Diversity-Enabled Sweet Spots for Robust Humanoid Locomotion". In: *arXiv:2510.14947*. Preprint. 2025.
- [9] Gilbert Feng, Hongbo Zhang, Zhongyu Li, Xue Bin Peng, Bhuvan Basireddy, Linzhu Yue, Zhitao Song, **Lizhi Yang**, Yunhui Liu, Koushil Sreenath, and Sergey Levine. "Genloco: Generalized Locomotion Controllers for Quadrupedal Robots". In: *Conference on Robot Learning (CoRL)*. PMLR, 2023, pp. 1893–1903.
- [10] Xiaoyu Huang, Zhongyu Li, Yanzhen Xiang, Yiming Ni, Yufeng Chi, Yunhao Li, **Lizhi Yang**, Xue Bin Peng, and Koushil Sreenath. "Creating a Dynamic Quadrupedal Robotic Goalkeeper with Reinforcement Learning". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2023.
- [11] Chong Zhang and **Lizhi Yang**. "Generating a Terrain-Robustness Benchmark for Legged Locomotion: A Prototype via Terrain Authoring and Active Learning". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2023.
- [12] ***Lizhi Yang**, *Zhongyu Li, Jun Zeng, and Koushil Sreenath. "Bayesian Optimization Meets Hybrid Zero Dynamics: Safe Parameter Learning for Bipedal Locomotion Control". In: *IEEE International Conference on Robotics and Automation (ICRA)*. 2022.
- [13] **Lizhi Yang**, Ruhang Ma, and Avidesh Zakhori. "Drone Object Detection Using RGB/IR Fusion". In: *Electronic Imaging: Computational Imaging*. 2022.
- [14] Chenyu Yang, Guo Ning Sue, Zhongyu Li, **Lizhi Yang**, Haotian Shen, Yufeng Chi, Akshara Rai, Jun Zeng, and Koushil Sreenath. "Collaborative Navigation and Manipulation of a Cable-Towed Load by Multiple Quadrupedal Robots". In: *IEEE Robotics and Automation Letters* 7.4 (2022), pp. 10041–10048.
- [15] Zixian Zang, Haotian Shen, **Lizhi Yang**, and Avidesh Zakhori. "Sensor-Aware SLAM-based Frontier Exploration and Mapping". In: *Electronic Imaging: AVM*. 2022.
- [16] *Scott Gilroy, *Derek Lau, ***Lizhi Yang**, Ed Izaguirre, Kristen Biermayer, Anxing Xiao, Mengti Sun, Ayush Agrawal, Jun Zeng, Zhongyu Li, and Koushil Sreenath. "Autonomous Navigation for Quadrupedal Robots with Optimized Jumping Through Constrained Obstacles". In: *IEEE CASE*. 2021.
- [17] *Anxing Xiao, *Wenzhe Tong, ***Lizhi Yang**, Jun Zeng, Zhongyu Li, and Koushil Sreenath. "Robotic Guide Dog: Leading a Human with Leash-Guided Hybrid Physical Interaction". In: *IEEE International Conference on Robotics and Automation (ICRA)*. Best Paper Award Finalist for Service Robotics. 2021.
- [18] **Lizhi Yang**, Ilian Herzi, Avidesh Zakhori, Anup Hiremath, Sahm Bazargan, and Robert Tames-Gadam. "Indoor Query System for the Visually Impaired". In: *International Conference on Computers Helping People with Special Needs (ICCHP)*. 2020, p. 517.
- [19] Huijuan Xu, **Lizhi Yang**, Stan Sclaroff, Kate Saenko, and Trevor Darrell. "Spatio-Temporal Action Detection with Multi-Object Interaction". In: *EPIC Workshop @ European Conference on Computer Vision (ECCV)*. 2020.

Invited Talks

2025: *From Performance to Safety: Bringing Robots Into the Real World*, Seminar of Prof. Xiaobin Xiong, University of Wisconsin–Madison; Seminar of Prof. Adam Wierman, California Institute of Technology

Academic Service

Reviewer: IEEE Transactions on Industrial Electronics (TIE), IEEE Robotics and Automation Letters (RA-L), IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) and Frontiers in Robotics and AI.

Outreach & Community Service

Pioneers in Engineering PiSens

Project Manager

Led team to develop sensor kits for low-cost robot competitions; organized annual competitions for underrepresented high school students in the Bay Area.

UC Berkeley

May 2019 – May 2020